PRIOR

Inspectable contribution maps for scientific synthesis

AIASci 2026

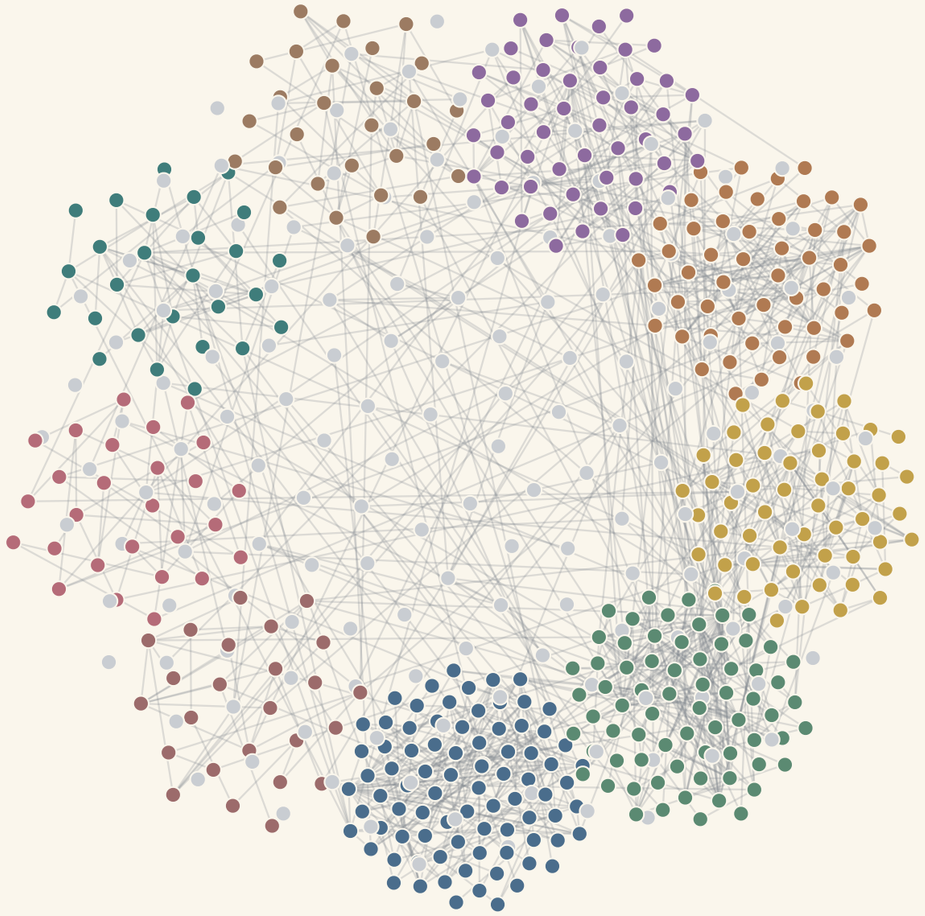
End-to-End Research Workflow, Methods, and Applications for AI-Assisted Science

San Francisco, USA

Klara Kaleb, Harit Vishwakarma, Callum Young, Yee Whye Teh

UNIVERSITY OF OXFORD

Leverhulme Centre for Nature Recovery



Agents4Academia

Built by researchers, for researchers

Oxford–Singapore Hackathon, June 2026

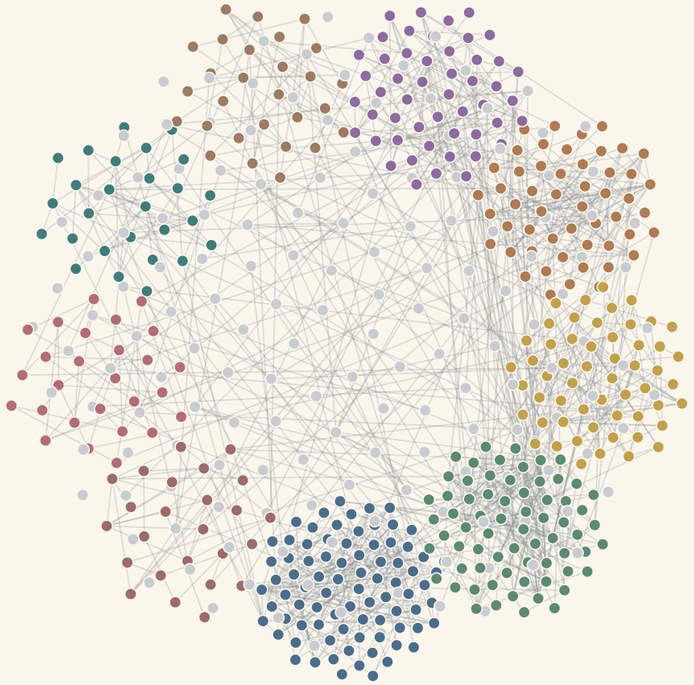
Oxford, UK

UNIVERSITY OF OXFORD

Singapore

NUS National University of Singapore

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE



Participants' top workflow frustration:

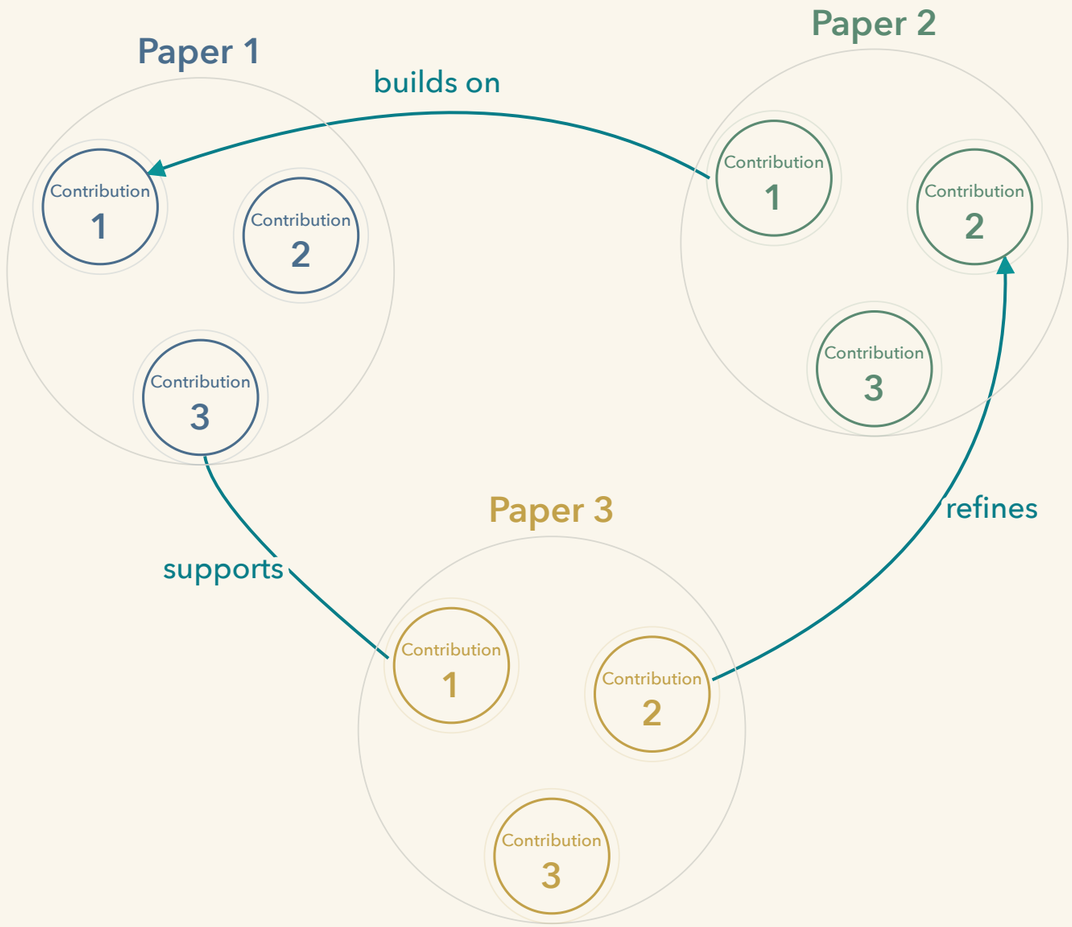
keeping up with the literature.

agents4academia.org

Hackathon supported by

ANTHROP\C

Modelling the literature beyond paper citations



A four-stage agentic workflow

Each new paper adds contributions and more relationships to check.
Agents repeat this reading and comparison across the corpus, retaining source evidence.

1

Scoper

Find and screen papers

Paper corpus

2

Contributor

Extract contributions

Statements + source quotations

3

Cartographer

Propose relationships

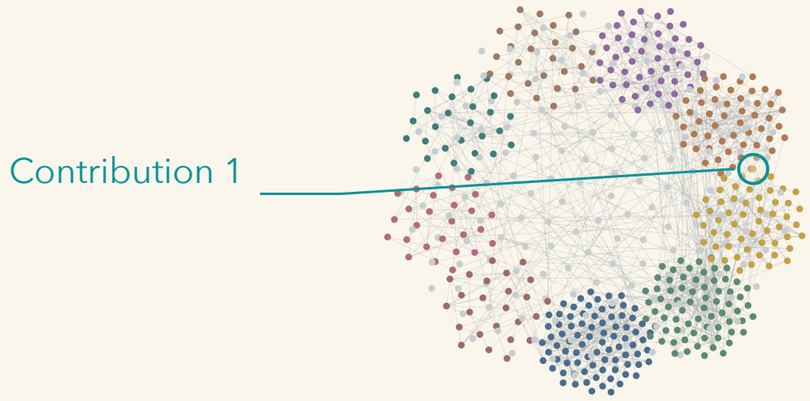
Typed links + evidence

4

Navigator

Inspect the map

Browsable contribution map



An AI-scientist research map

A fast-moving field: **73% of this corpus appeared in 2025–26.**
Systems combine retrieval, hypothesis generation, experimentation and review.

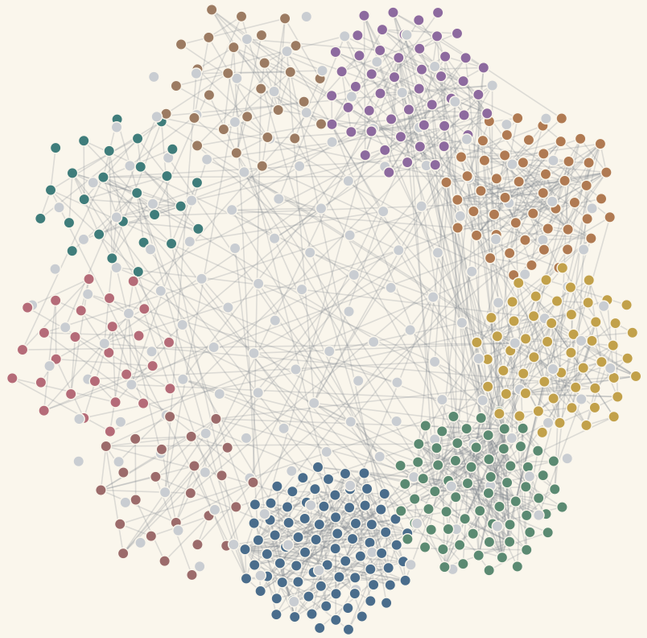
152
papers

581
author-declared contributions

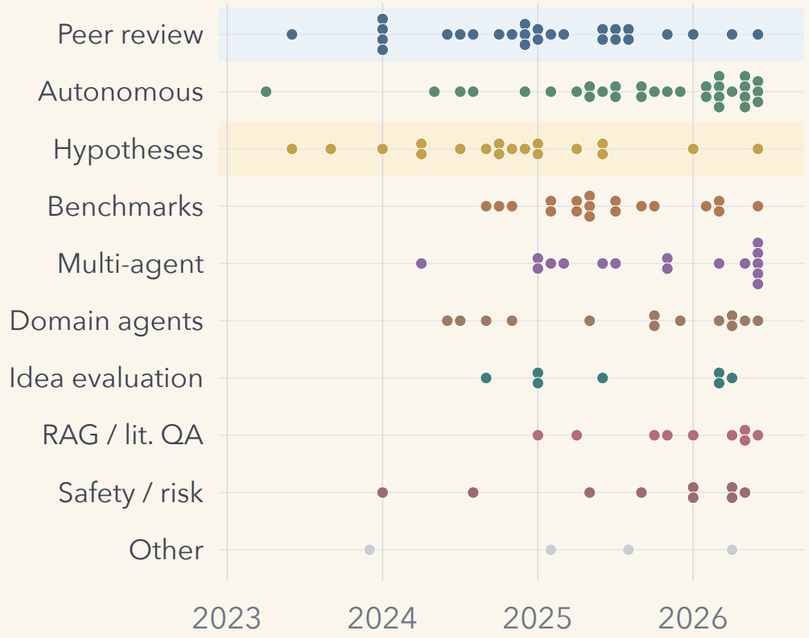
989
candidate links

> **55% of papers** span multiple displayed communities.

4 · Navigator: communities over time



Publication history



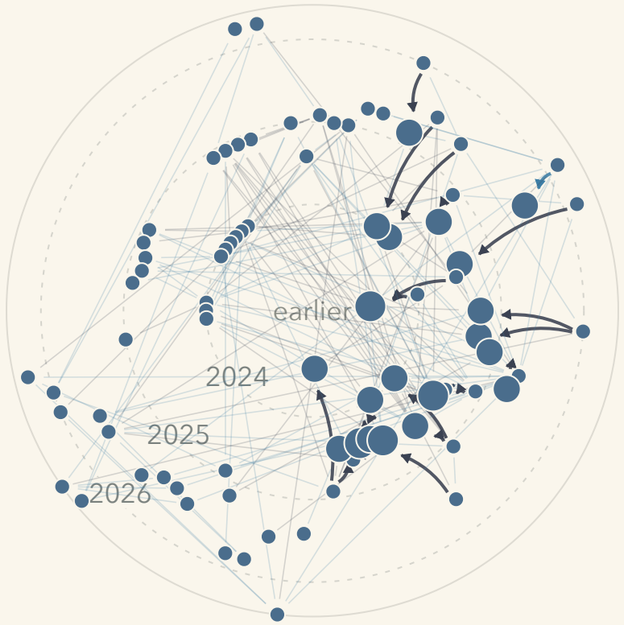
Peer review · Autonomous systems · Hypothesis generation · Benchmarks & eval · Multi-agent orchestration
Domain-science agents · Idea novelty / eval · RAG / literature-QA · Safety / risk · Other / unassigned

Concentrated and distributed reuse



Peer review

82 contributions



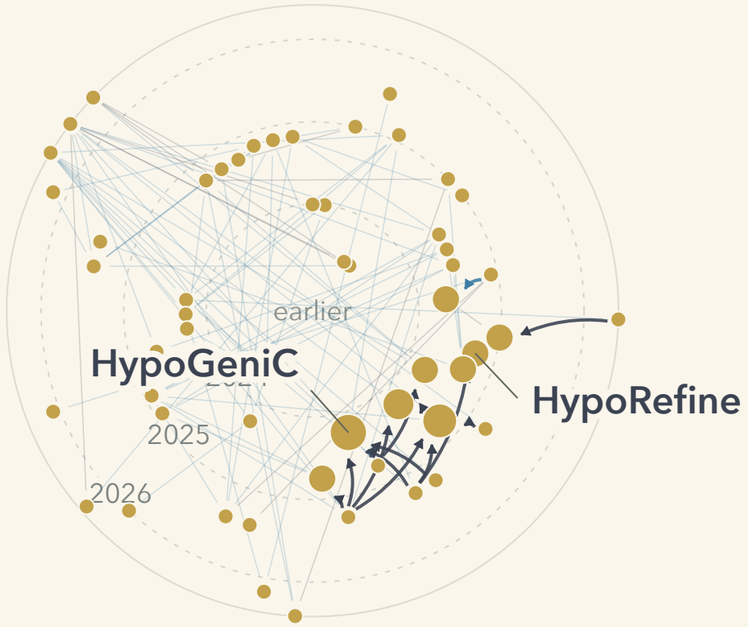


24 dependencies

across 13 target papers

Hypothesis generation

55 contributions





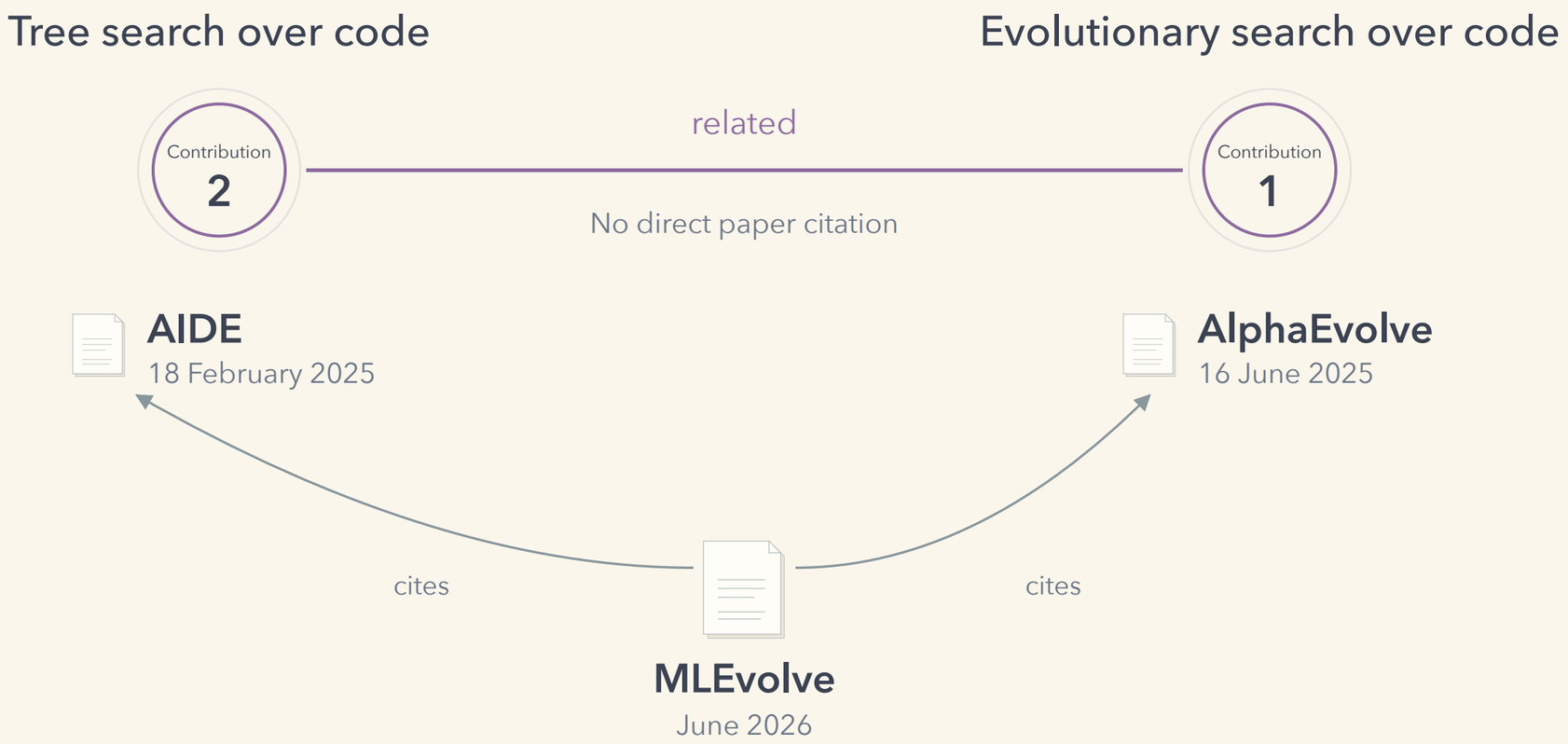
12 of 15 dependencies

target 2 papers

Figure 2: HypoGeniC: Zhou et al. (2024); HypoRefine: Liu et al. (2024).

A closer look at proposed connections

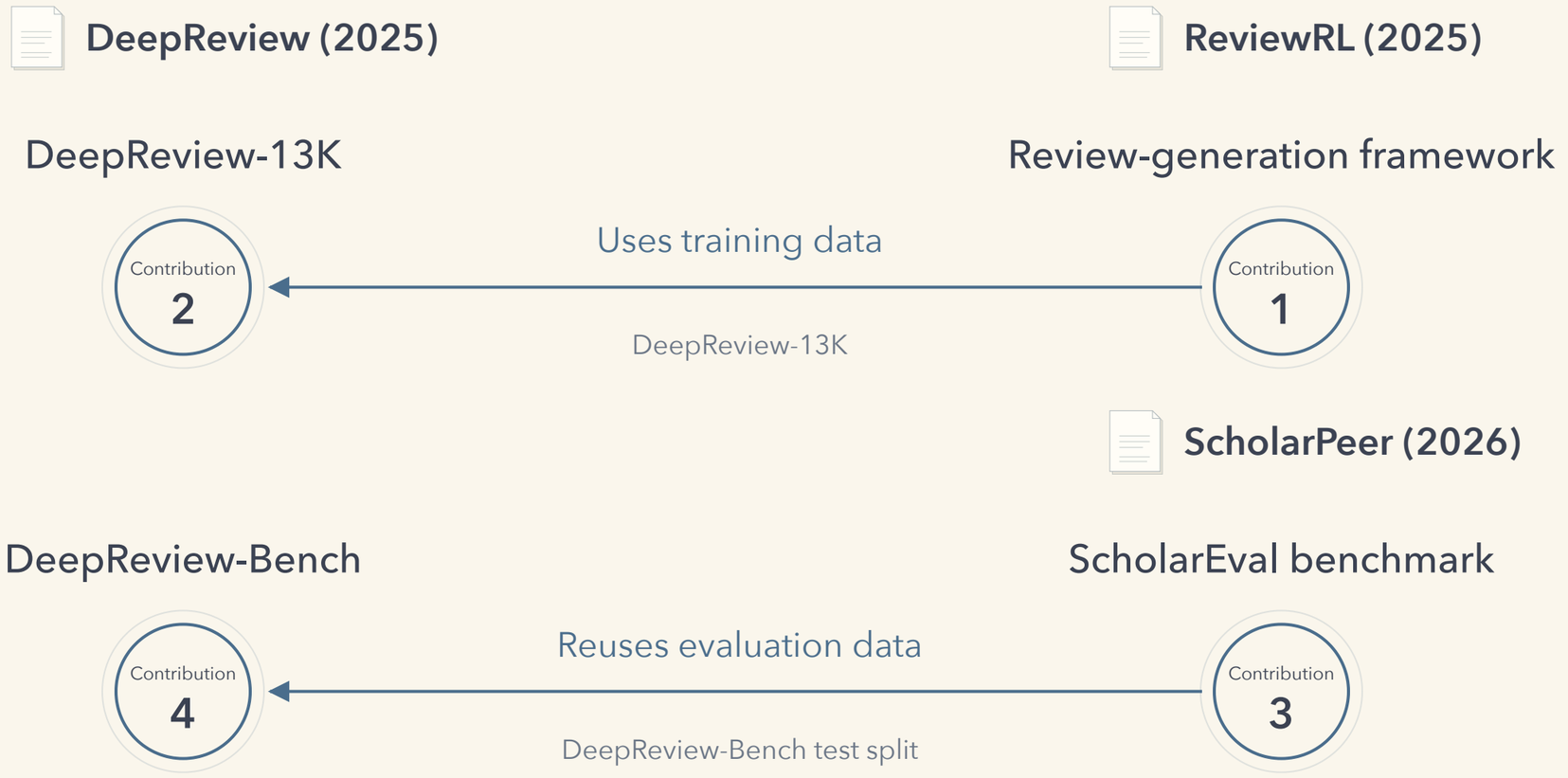
Example 1 · Code-solution search



Later co-cited **33%** PRIOR candidates 46 / 138 **23%** other uncited pairs 554 / 2,424

A closer look at proposed connections

Example 2 · Training and evaluation data



Future work

Further evaluation

Extraction fidelity, relationship correctness and usefulness for humans and agents performing research tasks

Across topics

Mathematics as a fast-moving next test case

A living evidence base

Add new papers; review, correct and update existing records.



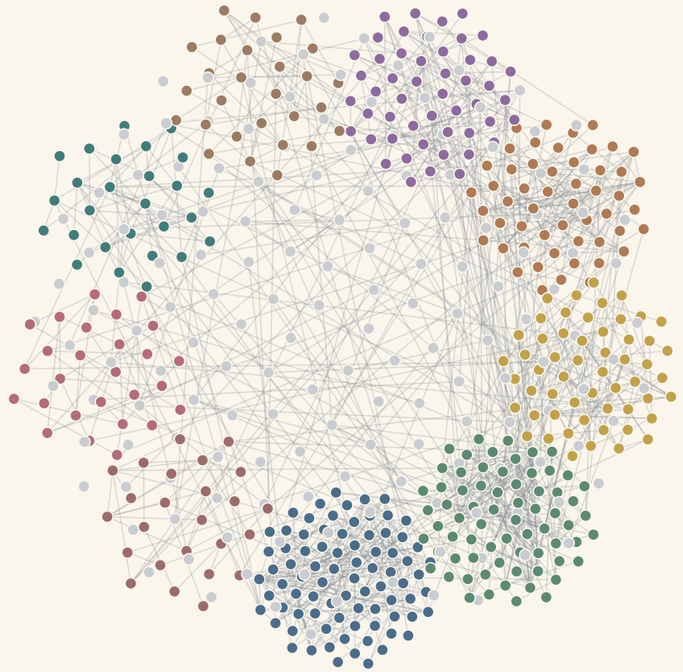
PRIOR

From a static snapshot
to a living evidence map

A four-stage agentic mapping workflow

An AI-scientist research map

askprior.ai



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Agents4Academia

A community of researchers building and understanding
agentic tools for academic work.

We're planning our next hackathon events.
Join us as a participant, collaborator or sponsor.

agents4academia.org

What the relation labels mean

Label	Reading of a proposed relation	Direction in the map
related	A substantive connection without a stronger established relation	Undirected
supports	Evidence or findings consistent with support for another contribution	Undirected
builds-on	Uses or adapts an earlier contribution	Dependent to reused
refines	Develops or modifies an earlier contribution	Dependent to reused
contradicts	Incompatible findings under sufficiently comparable conditions	Undirected
none / unclear	No relation proposed / insufficient basis for a clear judgment	No scientific link asserted